EN.530.685: Attitude Determination & Control

Instructor: Prof. Abhishek Cauligi

Semester: Spring 2027

Course website: acelab.me.jhu.edu/adcs

Course Description

In this course, students will learn the fundamentals of how a spacecraft attitude (i.e., orientation in inertial space) is determined and controlled. In particular, this course will study how to model the kinematics and dynamics of a rigid body in space, the representation of its attitude for different applications, how to use state estimation and sensor fusion techniques to estimate the attitude using noisy sensor measurements, and how to control the attitude using passive and active control techniques. Through the course of the semester, students will build up a spacecraft simulation framework using Google's `jax` scientific programming library and applications studied will include satellites, rockets, and quadrotors/drones. Students are expected to have a strong foundation in linear algebra and differential equations as well as experience with a high-level programming language (e.g., Python or Julia).

Key words: Spacecraft kinematics and dynamics, spacecraft modeling, state estimation, uncertainty quantification, linear and nonlinear attitude control.

Prerequisites: A strong foundation in linear algebra (e.g., `EN.530.616 Introduction to Linear Systems Theory`) and differential equations (e.g., `EN.530.761 (01) Mathematical Methods of Engineering 1`) and experience with a high-level programming language such as Python or Julia will be assumed. This course assumes otherwise assumes no background in optimization.

Learning Outcomes

At the end of the semester, students will be able to:

- Model the attitude dynamics of a spacecraft system.
- Estimate the spacecraft state (i.e., orientation, angular velocity) using noisy sensor measurements.
- Control satellite attitude using linear and nonlinear feedback control laws.
- Understand the first principles approaches to simulating spacecraft dynamics.
- Characterizing the uncertainty and disturbances experienced by a satellite based on its orbital parameters.

Instructor

Abhishek Cauligi
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Teaching Assistants

TBD

Lectures

TBD.

Office Hours

Office hours will begin in the second week of the semester.

Assignments

- **Homeworks:** There will be four homeworks through the course of the semester. These assignments will include both programming and theoretical components and are meant to give students practical experience implementing optimal control algorithms ahead of their final project.
- **Quizzes:** There will five short quizzes held the week after an assignment is due in class that are written, in-person, and closed note. These quizzes should take approximately $15 - 20$ minutes and are intended to provide students the opportunity to track their theoretical learning.
- **Final Project:** Students will work in project teams to complete a final project through the course of the semester. Further details will be provided in the final project guidelines.

Grading Policy

- Homeworks: 20%
- Quizzes: 45%
- Final Project: 30%
- Participation: 5%

Course Policies

Late Assignments: Late submissions will be penalized by 10% per day and the final deadline for consideration is two days after the assignment due date.

Attendance: Regular attendance is expected and in-class attendance will be taken.

Textbook

There is no required textbook for this class. However, the following resources may be useful:

- NocedalWright2006

Ethics

Academic Integrity: All students must adhere to university policies on plagiarism and cheating. Students are allowed to collaborate to complete problem sets, however all submitted material (including code, proofs, derivations, among others) must have been independently completed. If a student does indeed collaborate on a homework assignment, then they must include this detail and the nature of the collaboration as a part of the submitted assignment. Students can find further information on academic integrity policies for undergraduates here and for graduate students here. Any violations of these policies as determined by the instruction team will subsequently dealt with appropriately [1]

Course Policy on Generative AI: Students are permitted to use AI assistants (e.g., chatbots, large language models, etc.) to assist them in the competition of homework and final project assignments. However, students are not permitted to directly submit and claim as their own any material that is substantially copied over from the output of such an AI coding agent. This includes, but is not limited to, e.g., providing an AI assistant the homework files and asking the assistant to complete it. As such, any use of an AI assistant for the homework or final projects is expected to be reported in the same manner of collaboration as with another student in class is expected to be. Suspected breaches of this policy will be treated and addressed as violations of the aforementioned academic integrity standards.

In general, if there is any uncertainty or additional clarification necessary on academic integrity, ethics, and honesty guidelines, then please ask the instructor.

References

[1] As written on October 12, 2026, the academic misconduct policy details can be found here.